Topic 2 Review Test · Section 504

Algebra I · Module 2, Topic 2: Transforming and Comparing Linear Functions · Monday, Nov 2

Name _____ Period _____ Date _____

Directions: This review has the same kinds of questions as the Topic 2 test. Read each question carefully. Show your work. **Select TWO** means choose exactly two answers. Finish Part 1 first; Part 2 is extra practice.
One question is on each page. Take your time. You may use a calculator and the reference box. Ask your teacher if you need a break or a question read aloud.

Reference box

Slope: $\mathbf{m = (y_2 - y_1) \div (x_2 - x_1)}$ · Slope-intercept form: $\mathbf{y = mx + b}$ · Point-slope form: $\mathbf{y - y_1 = m(x - x_1)}$

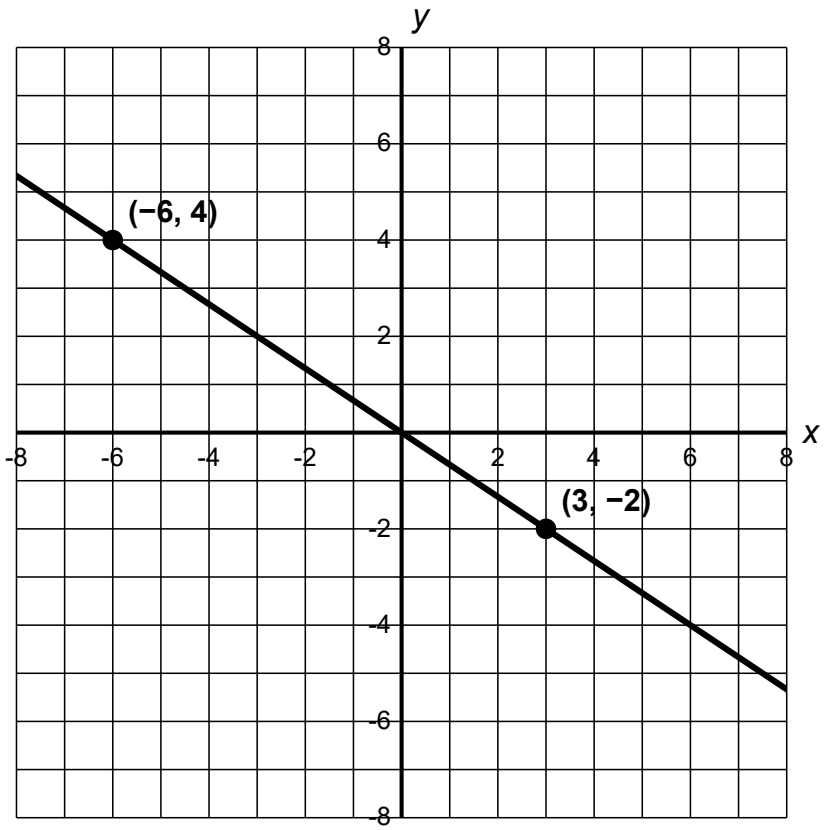
Parallel lines: same slope. **Perpendicular** lines: negative reciprocal slopes (2 and −½).

f(x) + k: up k · **f(x) − k**: down k · **f(x − h)**: right h · **f(x + h)**: left h · **a·f(x)** or **f(ax)**: steeper or flatter

Domain = inputs (x). **Range** = outputs (y). **Zero** = where the line crosses the x-axis.

Part 1 · Just like the test (Questions 1–10)

1. The graph of linear function m is shown on the grid.



Given $f(x) = x$ and $m(x) = af(x)$, what is the value of a?

$a =$ _____

2. The cost of renting a bounce house, y, for x hours at a birthday party can be modeled using a linear function. The rental company charges a fixed fee of $75, plus an additional $20 per hour. The bounce house can be rented for a maximum of 5 hours per day. For one day of rental, what is the **range** of the function for this situation?

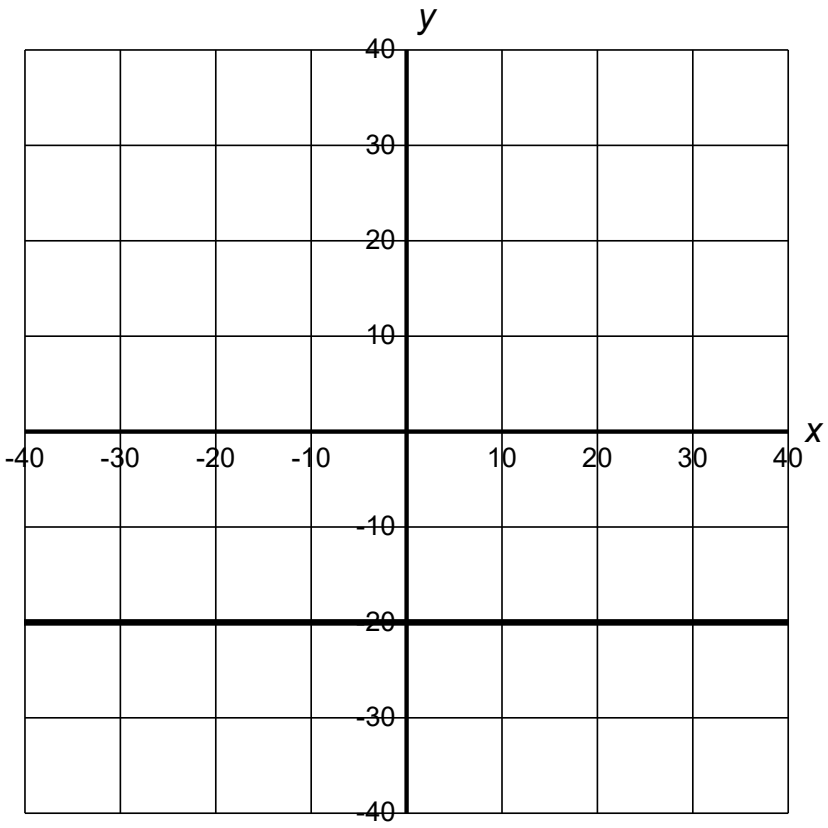
A. $0 \leq x \leq 5$

B. $20 \leq y \leq 175$

C. $75 \leq y \leq 175$

D. $0 \leq x \leq 7$

3. What are the **equation and slope** of the line shown on the grid?



F. $x = -20$; slope is zero

G. $y = -20$; slope is undefined

H. $y = -20$; slope is zero

J. $x = -20$; slope is undefined

4. Marcus graphed $f(x)$ and $g(x) = f(x - 3)$ on the same coordinate grid. Which statement describes how the graphs of f and g are related?

A. The graph of the function $f(x)$ is translated up 3 units to produce $g(x)$.

B. The graph of the function $f(x)$ is translated down 3 units to produce $g(x)$.

C. The graph of the function $f(x)$ is translated right 3 units to produce $g(x)$.

D. The graph of the function $f(x)$ is translated left 3 units to produce $g(x)$.

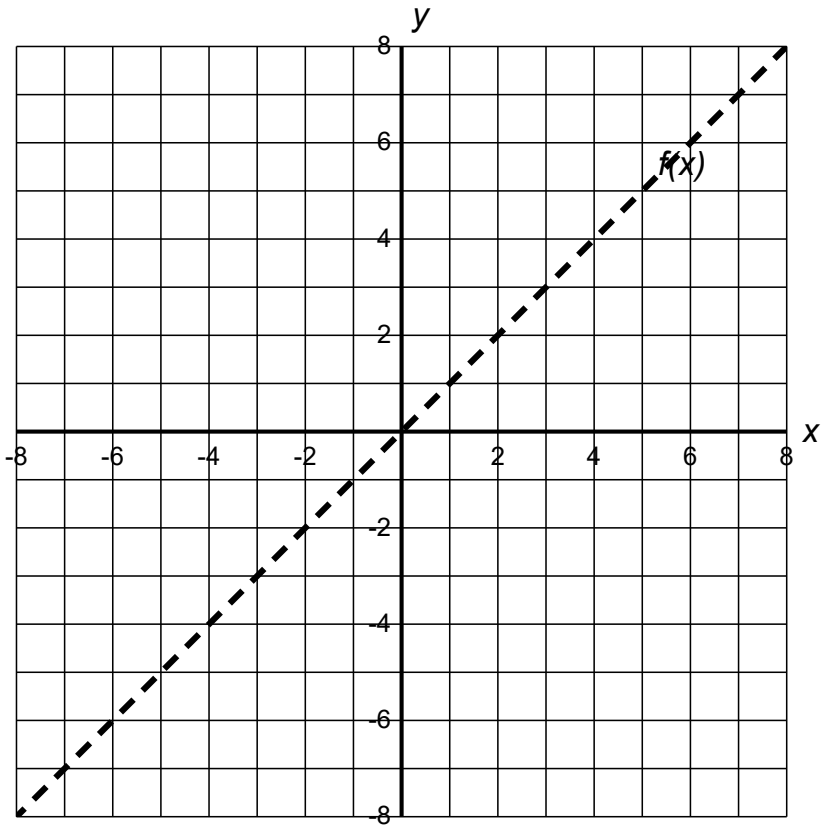
5. What is the ***y*-intercept** of the linear function created by translating $f(x)$, represented in the table, **2 units right**?

x	$f(x)$
−2	−7
−1	−4
0	−1
1	2
2	5
3	8

y-intercept = _____

6. Graph $g(x)$ based on the given transformation of $f(x)$.

$f(x) = x$ $\quad g(x) = f(2x)$



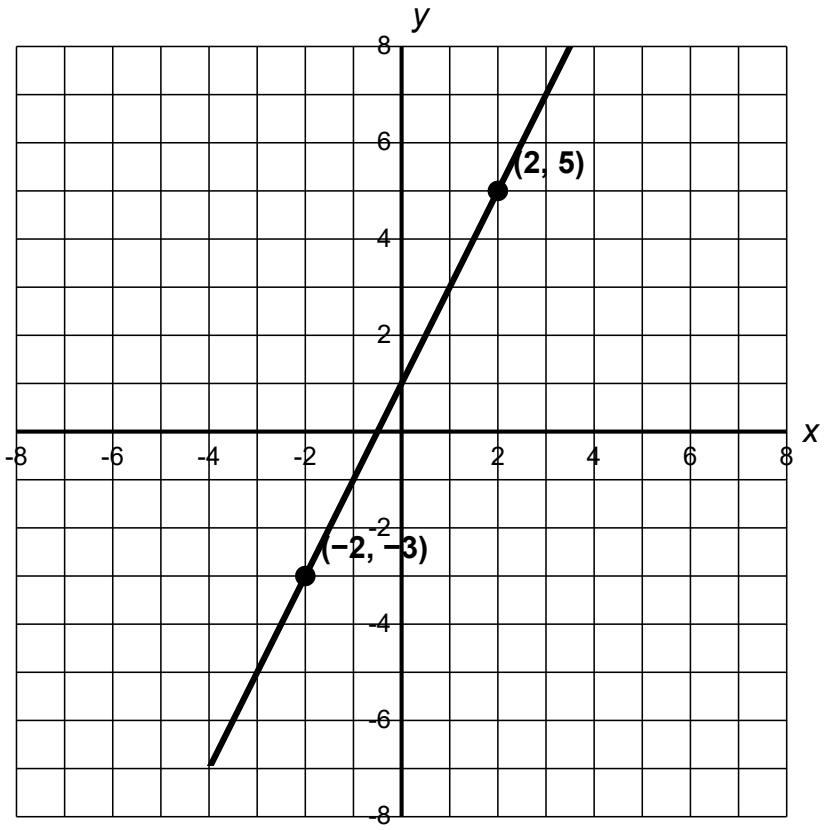
7. What is the equation in **slope-intercept form** of the line that crosses the *x*-axis at −6 and is **perpendicular** to the line represented by $y = \frac{3}{2}x - 1$?

Equation: _____

8. What is the equation in **slope-intercept form** of the line that passes through the point (5, 0) and is **parallel** to the line represented by $y = -2.4x + 1.6$?

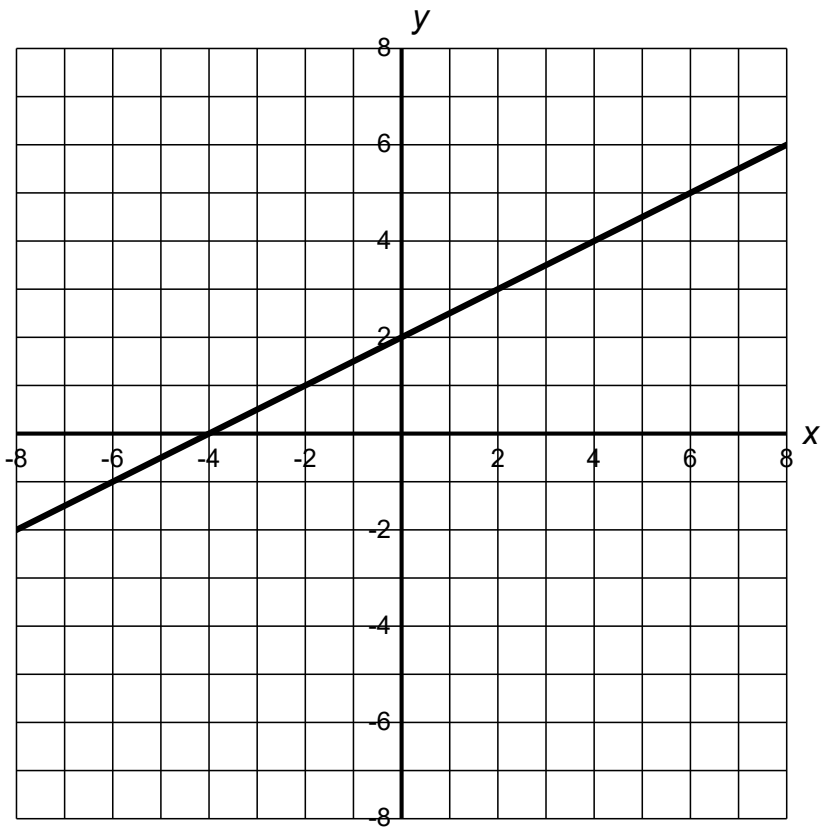
Equation: _____

9. Write an equation in **point-slope form** that represents the graph.



Equation: _____

10. The graph of a linear function is shown on the coordinate plane.



Which statements describe the graph of the linear function? **Select TWO correct answers.**

F. The graph is increasing.

G. The slope of the line is 2.

H. The *y*-intercept is (0, −4).

J. The zero of the function is −4.

K. The zero of the function is 2.

Part 2 · Extra practice (Questions 11–14)

11. Let $f(x) = x - 1$, $g(x) = f(2x)$, and $h(x) = 2f(x)$. Which statement is **true**?

A. $g(x)$ and $h(x)$ are the same function.

B. $g(x)$ and $h(x)$ have the same slope but different y-intercepts.

C. $g(x)$ and $h(x)$ have different slopes but the same y-intercept.

D. $g(x)$ and $h(x)$ have different slopes and different y-intercepts.

12. Which pair of equations represents **perpendicular** lines?

F. $y = 5x + 1$ and $y = -5x + 2$

G. $y = -\frac{1}{3}x + 4$ and $y = 3x - 2$

H. $y = \frac{4}{5}x$ and $y = \frac{5}{4}x - 1$

J. $y = 2x + 3$ and $y = 2x - 3$

13. $f(x) = 2x + 1$ and $g(x) = f(x) - 4$. Write the equation of $g(x)$ and **describe how the graph** of f moves to make the graph of g.

$g(x) =$ _____

Description: _____

14. A tank holds 300 gallons of water. It drains at 15 gallons per minute until it is empty. Let x be the number of minutes. What is the **domain** of the function for this situation?

A. $0 \leq x \leq 300$

B. $0 \leq x \leq 20$

C. $0 \leq y \leq 300$

D. $15 \leq x \leq 300$